
Project Proposal: Centralized University Portal & Placement Management System (CUP-PMS)

Target Institution: University of Hyderabad (UoH)

Team Members: 22MCCE13 & 22MCCE16

1. Introduction / Problem Statement

Universities increasingly struggle with fragmented placement processes and siloed systems. At present, the Training and Placement Office (TPO) maintains student records, company eligibility criteria, and interview schedules manually using disparate tools like spreadsheets and ad-hoc emails. This traditional, "manpower-intensive" approach leads to delays in communication, data redundancy, and coordination errors during peak hiring seasons.

The University of Hyderabad (UoH) requires a unified, data-driven web portal that seamlessly connects students, academic faculty, administrators, and corporate recruiters. This project addresses the current gap by proposing a centralized, role-based platform that automates critical workflows—such as student eligibility verification, resume distribution, and real-time interview notifications—transforming the placement lifecycle into an efficient digital ecosystem.

2. Objectives of the Project

- **Unified Access & Identity:** Develop a secure, Single Source of Truth (SSOT) portal linking all stakeholders using UoH Permanent Registration Numbers (PRNs) and Role-Based Access Control (RBAC).
- **Workflow Automation:** Eliminate manual data parsing by implementing an "Intelligent Eligibility Engine" that automatically shortlists students based on recruiter-defined criteria (e.g., CGPA thresholds, branch constraints, and active backlogs).
- **Real-Time Communication:** Replace delayed email chains with instant push notifications and alerts for interview scheduling and placement status updates.
- **Integrated Assessments:** Provide an internal "Quiz Designer" module for recruiters and TPOs to conduct preliminary technical assessments directly on the platform.
- **Actionable Analytics:** Generate real-time visual dashboards and exportable reports for the administration to track departmental placement metrics, average Cost to Company (CTC), and hiring trends.

3. Project Scope & Core Modules

The system will be bounded by the following core modules:

1. **Authentication & Security:** Utilizing **Firebase Auth** for secure SSO, Multi-Factor Authentication (MFA), and JWT-based session management.
2. **Student Profile & Document Management:** Students can build digital profiles and upload resumes (PDFs) securely hosted via **Firebase Cloud Storage**, with strictly controlled visibility.
3. **Recruitment & Job Management:** Recruiters can post detailed job descriptions (Role, CTC, Deadlines). The system cross-references these posts with the PostgreSQL student database to enforce eligibility rules dynamically.
4. **Assessment Engine:** A timed, MCQ-based testing environment for initial candidate screening.
5. **Notification Hub:** Automated alert system utilizing **Firebase Cloud Messaging (FCM)** for real-time interview coordination and SMS/Email fallbacks via external APIs (e.g., Twilio/SendGrid).

4. Proposed System Architecture

The project will utilize a **Three-Tier Hybrid Cloud Architecture**, specifically chosen to balance ACID-compliant data integrity with real-time cloud capabilities.

- **Presentation Layer (Frontend):** Built using **React.js / Next.js**, providing a responsive, mobile-first Single Page Application (SPA) with distinct dashboards for Students, Recruiters, and Admins.
- **Application Layer (Backend):** A stateless **Node.js / Express** REST API that handles business logic, security middleware, and data processing.
- **Data & Persistence Layer:**
 - * **PostgreSQL:** Serving as the primary relational database to handle complex queries, academic records, and transactional placement data ensuring high data integrity.
 - **Firebase Services:** Acting as the real-time and media layer (Auth, Storage, and Messaging) to reduce server load and ensure high availability.

5. Expected Outcomes and Benefits

- **For Students:** A transparent, mobile-friendly interface to track applications, manage resumes, take assessments, and receive instant interview updates without relying on fragmented WhatsApp groups.
- **For Recruiters:** A powerful filtering dashboard to effortlessly parse through thousands of profiles, shortlist eligible candidates in one click, and manage interview rounds efficiently.
- **For the TPO & Administration:** Massive reduction in administrative overhead, elimination of manual data entry errors, and instant access to verifiable placement statistics for university accreditations.
- **For the University:** Enhanced institutional reputation through a professional corporate

interface and strict adherence to data privacy standards (AES-256 encryption and HTTPS).

6. Conclusion

This proposal outlines a highly scalable, modern placement portal tailored for the University of Hyderabad. By leveraging an iterative Agile SDLC and a robust tech stack (React, Node, PostgreSQL, Firebase), the project will deliver an enterprise-grade platform. Detailed Software Requirements Specifications (SRS), UML diagrams, and API documentation will govern the development process, ensuring the final product successfully transitions UoH from a disjointed manual process to a streamlined, secure, and fully automated recruitment engine.